

Case Study 6 — Detecting an Unusual Customer

Question

A shopping website finds that the average amount spent by customers is ₹2,000, with a standard deviation of ₹400.

A particular customer spends ₹3,200.

Questions:

1. Calculate the customer's z-score.
2. Is the customer's spending unusually high compared with the average?
3. What would a negative z-score mean in this context?
4. If another customer has a z-score of -1.5 , what does it tell you?

Step-by-Step Solution

Given: Mean $\mu = 2000$, Standard deviation $\sigma = 400$, Customer spending $x = 3200$.

Step 1: Use the z-score formula $z = (x - \mu) / \sigma$.

Step 2: Substitute the values $z = (3200 - 2000) / 400$.

Step 3: Simplify $1200 / 400 = 3$.

Answer: Z-score = 3.

Interpretation: The customer spends 3 standard deviations above the average, so the spending is unusually high.

A negative z-score means the spending is below the average amount.

If $z = -1.5$, the customer spends 1.5 standard deviations below the average, which indicates lower-than-average spending but not necessarily an extreme outlier.

Conclusion: Z-scores help businesses identify unusual customer spending behaviour and compare individual customers with the overall population.

Case Study 7 — Delivery Time Analysis

Question

Delivery times (minutes): 25, 27, 28, 29, 30, 30, 31, 32, 34, 60.

Questions:

1. Calculate the mean and median.
2. Find Q1 and Q3.
3. Calculate the IQR.
4. Check for outliers using the $1.5 \times \text{IQR}$ rule.
5. Explain whether the mean or median gives a better description of the typical delivery time.

Step-by-Step Solution

Step 1: Add all delivery times: $25 + 27 + 28 + 29 + 30 + 30 + 31 + 32 + 34 + 60 = 326$.

Step 2: Mean = $326 / 10 = 32.6$ minutes.

Step 3: Median = average of the 5th and 6th values = $(30 + 30) / 2 = 30$ minutes.

Step 4: Lower half = 25, 27, 28, 29, 30, so Q1 = 28.

Step 5: Upper half = 30, 31, 32, 34, 60, so $Q3 = 32$.

Step 6: $IQR = Q3 - Q1 = 32 - 28 = 4$.

Step 7: $1.5 \times IQR = 1.5 \times 4 = 6$.

Lower limit = $28 - 6 = 22$, Upper limit = $32 + 6 = 38$.

Since 60 is greater than 38, 60 is an outlier.

The median is a better measure because the outlier 60 increases the mean and makes it less representative of typical delivery times.

Case Study 8 — Two Cities' Temperatures

Question

City A: 28, 29, 30, 30, 31, 30, 32.

City B: 20, 25, 30, 35, 30, 25, 45.

Questions:

1. Calculate the mean temperature for each city.
2. Calculate the variance for each city.
3. Which city has greater temperature variability?
4. Why can two datasets have the same mean but different standard deviations?
5. Which city has more predictable temperatures?

Step-by-Step Solution

Step 1: Sum of City A temperatures = 210.

Step 2: Sum of City B temperatures = 210.

Step 3: Mean for both cities = $210 / 7 = 30^{\circ}\text{C}$.

Step 4: Calculate squared deviations for each city.

City A variance ≈ 1.43 .

City B variance ≈ 57.14 .

Because City B has a much larger variance, it has greater temperature variability.

Two datasets can have the same mean but different spreads because the values may be tightly clustered or widely dispersed around the mean.

City A is more predictable because its temperatures stay close to 30°C throughout the week.

Case Study 9 — Data Center CPU Usage

Question

CPU usage percentages: 40, 42, 45, 45, 46, 48, 50, 52, 55, 90.

Questions:

1. Calculate mean, median, and mode.
2. Calculate the range.
3. Find Q1, Q3 and IQR.
4. Identify possible outliers.
5. Determine the direction of skewness.

Step-by-Step Solution

Step 1: Add all values = 513.

Step 2: Mean = $513 / 10 = 51.3$.

Step 3: Median = $(46 + 48) / 2 = 47$.

Step 4: Mode = 45 because it appears most frequently.

Step 5: Range = $90 - 40 = 50$.

Step 6: Q1 = 45, Q3 = 52.

Step 7: $IQR = 52 - 45 = 7$.

Step 8: $1.5 \times IQR = 10.5$, so the upper limit is $52 + 10.5 = 62.5$.

Since 90 is greater than 62.5, 90 is an outlier.

The very high value 90 pulls the distribution to the right, so the data is positively skewed.

Case Study 10 — Comparing Two Datasets

Question

Dataset A: 10, 20, 30, 40, 50.

Dataset B: 28, 29, 30, 31, 32.

Questions:

1. Calculate the mean of both datasets.
2. Calculate the variance of both.
3. Calculate the standard deviation of both.
4. Which dataset has greater spread?
5. If both datasets represent model errors, which model appears more consistent?
6. Explain why mean alone is not enough to compare datasets.

Step-by-Step Solution

Step 1: Mean of Dataset A = $(10 + 20 + 30 + 40 + 50) / 5 = 30$.

Step 2: Mean of Dataset B = $(28 + 29 + 30 + 31 + 32) / 5 = 30$.

Step 3: Squared deviations for Dataset A = 400, 100, 0, 100, 400.

Variance of Dataset A = $1000 / 5 = 200$.

Step 4: Squared deviations for Dataset B = 4, 1, 0, 1, 4.

Variance of Dataset B = $10 / 5 = 2$.

Step 5: Standard deviation of Dataset A = $\sqrt{200} \approx 14.14$.

Step 6: Standard deviation of Dataset B = $\sqrt{2} \approx 1.41$.

Dataset A has much greater spread because its standard deviation is much larger.

Dataset B is more consistent because its values stay very close to the mean.

Mean alone is not enough because two datasets can have the same centre but very different spread, variability, and presence of outliers.

